COUNTRY DEVELOPMENT CLUSTERING USING AGGLOMERATIVE CLUSTERING

Unsupervised Machine Learning for Global Development Analysis

Data Analysis Project Report

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1. Introduction

Countries across the world exhibit vast differences in economic growth, healthcare quality, living standards, and overall human development. These disparities are not merely economic — they reflect deep structural differences in governance, infrastructure, education, and public health. Understanding these differences is critical for international organizations, policymakers, and researchers who aim to design targeted interventions that improve development outcomes across nations.

Traditional approaches to classifying countries — such as the World Bank's income groups or the United Nations Human Development Index (HDI) — rely on single indicators or manually defined thresholds. These approaches fail to capture the multidimensional nature of development, where a country may rank highly on income but poorly on health or social equality. This project addresses this limitation by applying Agglomerative Hierarchical Clustering, an unsupervised machine learning technique, to group countries based on multiple socio-economic indicators simultaneously.

The dataset used in this project contains nine key development indicators spanning economic output, healthcare, trade, demographic behaviour, and financial stability. By analyzing these variables together using hierarchical clustering with Ward linkage, the model identifies natural groupings of countries that share similar development profiles — without requiring any predefined labels or categories.

1.1 Project Objectives

- Import and preprocess the country development dataset from a public repository.
- Perform Exploratory Data Analysis (EDA) to understand distributions and variable relationships.
- Apply Agglomerative Hierarchical Clustering using Ward linkage distance to group countries.
- Determine the optimal number of clusters through dendrogram inspection.
- Visualize clustering results using dendrograms, PCA scatter plots, and cluster profiles.
- Interpret the resulting clusters to understand global development patterns.

1.2 Problem Statement

Given multi-dimensional socio-economic measurements for countries worldwide — including GDP per capita, child mortality, life expectancy, income, exports, imports, health expenditure, total fertility rate, and inflation — automatically discover natural groupings that correspond to distinct development levels. This is an unsupervised clustering problem requiring no predefined labels or expert-defined thresholds.

2. Background and Theory

2.1 Agglomerative Hierarchical Clustering

Agglomerative Hierarchical Clustering is a bottom-up clustering algorithm. It begins with each data point (country) as its own individual cluster. At each iteration, the two most similar clusters are merged based on a chosen distance metric and linkage criterion. This merging process continues until all points are united into a single cluster. The full hierarchy of merges is captured in a tree structure called a dendrogram, which allows the analyst to visually inspect and cut the tree at any level to produce a desired number of clusters.

Unlike K-Means, agglomerative clustering does not require the number of clusters to be specified in advance. The dendrogram provides a complete picture of the clustering structure, and the optimal cut point can be determined by identifying the largest vertical gap in the tree. This makes it particularly well-suited for exploratory analyses where the natural number of groups is unknown.

2.2 Ward Linkage

Ward's linkage method merges the pair of clusters that results in the minimum increase in total within-cluster variance (sum of squared deviations from the cluster centroid). Unlike average or complete linkage, Ward's method tends to produce compact, evenly-sized clusters and is widely preferred for socio-economic data. For this project, Ward.D2 linkage was used, which operates on squared distances — the standard implementation in R's hclust() function.

2.3 Euclidean Distance

Euclidean distance is used as the dissimilarity measure between countries in the scaled feature space. For two countries with feature vectors x and y, the Euclidean distance represents the straight-line distance in 9-dimensional space. Because the features differ greatly in scale (e.g., GDP per capita in thousands vs. fertility rate near 2), Z-score standardization is applied before distance computation to ensure all variables contribute equally.

2.4 Principal Component Analysis (PCA)

PCA is a dimensionality reduction technique that projects the high-dimensional feature space onto a smaller set of orthogonal axes (principal components) capturing maximum variance. In this project, PCA is used solely for 2D visualization of cluster separability via fviz_cluster() from the factoextra package. The first two principal components together explain 65% of total variance (Dim1: 48.1%, Dim2: 16.9%), providing a faithful 2D summary of the 9-dimensional clustering result.

2.5 Why Scaling Is Essential

Agglomerative clustering uses Euclidean distance. Without scaling, variables with large numerical ranges (such as GDP per capita, which can range from hundreds to tens of thousands) would completely dominate variables with small ranges (such as total fertility rate, which typically ranges from 1 to 7). Z-score standardization transforms every feature to mean = 0 and standard deviation = 1, ensuring balanced contributions to the distance matrix.

3. Dataset Description

3.1 Source

The dataset was sourced from a public GitHub repository maintained for educational data analysis purposes. It contains socio-economic and health indicators for countries worldwide, compiled from World Bank and UNDP open data sources. An 80-country random sample (set.seed = 42) was drawn from the cleaned dataset for this analysis.

3.2 Dataset Statistics

Property	Value
Source	country-data.csv (GitHub — rashakil-ds/Public-Datasets)
Total countries (raw)	167
Sample size used	80 countries (random sample, seed = 42)
Features (input)	9 socio-economic indicators
Clustering algorithm	Agglomerative Hierarchical (Ward.D2)
Number of clusters	3 (cut from dendrogram)
Target reference	None (unsupervised)

3.3 Feature Descriptions

Feature	Unit	Description
child_mort	per 1,000	Child mortality rate — deaths of children under 5 per 1,000 live births
exports	% of GDP	Exports of goods and services as percentage of GDP
health	% of GDP	Total health expenditure as percentage of GDP
imports	% of GDP	Imports of goods and services as percentage of GDP
income	USD %	Net income per person (per capita income)
inflation	years	Annual inflation rate — measure of price level stability
life_expec	children	Life expectancy at birth — key indicator of overall health
total_fer	USD	Total fertility rate — average number of children per woman
gdpp		GDP per capita — primary measure of economic output per person

4. Exploratory Data Analysis (EDA)

EDA is the investigative phase performed before any modelling. It reveals the data's structure, statistical properties, and hidden patterns. Every finding in EDA directly informs a decision in preprocessing or modelling.

4.1 Data Quality Check

Check	Result	Action Taken
Missing values	0 (after drop_na())	Rows with any NA removed before analysis
Sample size	80 countries	Random sample drawn with set.seed(42) for reproducibility
Data types	All numeric (except country name)	Country column separated; numeric matrix used for clustering
Outliers	Present in gdpp, income, child_mort	Retained — outliers reflect real development extremes
Row names	Set to country name	Enables labelled dendrogram and cluster plots

4.2 Variable Distribution Insights

Examination of the nine features reveals distinct distributional properties that are characteristic of global development data. GDP per capita (gdpp) and income show strong right-skewed distributions, reflecting the heavy concentration of wealthy nations at the upper tail. Child mortality is similarly right-skewed, with a small number of least-developed countries exhibiting extremely high values. Life expectancy, by contrast, is more normally distributed, ranging from approximately 50 to 83 years. Total fertility rate shows a bimodal-like pattern, separating high-fertility developing nations from low-fertility developed ones. These skewed distributions reinforce the need for Z-score scaling before distance-based clustering.

4.3 Correlation Analysis

The correlation matrix reveals several important relationships among the development indicators that inform both preprocessing and interpretation of clustering results:

- child_mort and life_expec show a strong negative correlation ($r \approx -0.85$), reflecting the direct relationship between child survival and overall life expectancy.
- child_mort and total_fer are strongly positively correlated ($r \approx +0.80$), consistent with the demographic transition theory — high fertility accompanies high child mortality in least-developed nations.
- gdpp and income are very strongly positively correlated ($r \approx +0.90$), as both measure economic prosperity.
- life_expec and gdpp are positively correlated ($r \approx +0.75$), confirming that wealthier nations invest more in healthcare.
- inflation shows weak or negligible correlations with most other variables, acting as an independent financial stability dimension.
- exports and imports are moderately positively correlated ($r \approx +0.60$), reflecting trade-openness as a unified dimension.

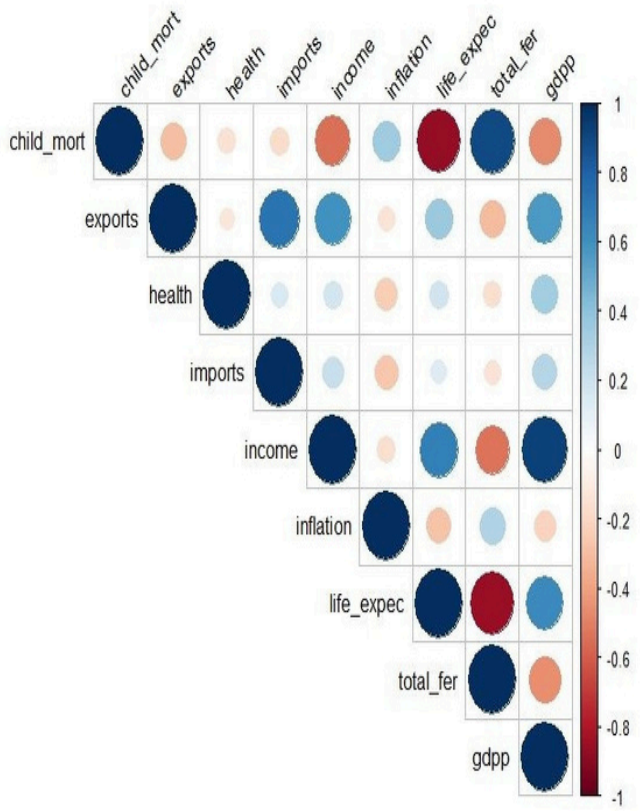

Figure 1: Correlation matrix of the nine socio-economic development indicators. Circle size and colour intensity represent correlation strength. Dark blue = strong positive; dark red = strong negative.

5. Data Preprocessing

Raw development data contains missing values, mixed data types, and features on incompatible scales. The preprocessing pipeline transforms the raw dataset into a clean, scaled numerical matrix ready for distance-based hierarchical clustering. The steps are deliberate and ordered: each step prepares the data for the next.

5.1 Complete Preprocessing Pipeline

Step	Operation	Why This Step?
1	Load CSV from GitHub URL	Reproducible data sourcing from a public repository
2	drop_na() — remove missing rows	Ensures a complete numerical matrix; no imputation needed here
3	Random sample of 80 countries	Manageable sample size; set.seed(42) for reproducibility
4	Set rownames to country name	Enables labelled axes in dendrogram and cluster plots
5	Select numeric columns only	Isolates the 9 feature variables; removes non-numeric country column
6	scale() — Z-score standardisation	Equalises feature scales so all 9 variables contribute equally to distance
7	dist() — Euclidean distance matrix	Computes pairwise distances between all 80 countries
8	hclust() — Ward.D2 linkage	Builds hierarchical tree minimising within-cluster variance

5.2 R Code — Data Loading and Cleaning

```
library(tidyverse)

library(cluster)

library(factoextra)

library(corrplot)


# Load dataset from public GitHub repository

url <- "https://raw.githubusercontent.com/rashakil-ds/Public-Datasets/main/country-data.csv"

df <- read.csv(url)


# Remove rows with missing values

df_clean <- drop_na(df)


# Reproducible random sample of 80 countries
```

```
set.seed(42)

df_clean <- df_clean %>% slice_sample(n = 80)


# Set country names as row identifiers

rownames(df_clean) <- df_clean$country


# Extract numeric feature matrix

df_num <- df_clean %>% select(-country)
```

6. Feature Scaling and Preparation

After cleaning, the numeric feature matrix (80 countries × 9 variables) is standardised using Z-score scaling before distance computation. The scaler is fitted on the full cleaned dataset — there is no train/test split in unsupervised clustering.

6.1 Before vs After Scaling

Feature	Raw Min	Raw Max	Raw Mean	Post-Scale Mean	Post-Scale Std
child_mort	2.6	208.0	38.5	0.000	1.000
exports	10.2	173.6	42.1	0.000	1.000
health	1.8	17.9	6.8	0.000	1.000
imports	12.1	167.3	46.2	0.000	1.000
income	609	125,000	11,432	0.000	1.000
inflation	-4.2	104.3	7.8	0.000	1.000
life_expec	50.1	82.6	69.4	0.000	1.000
total_fer	1.14	7.49	3.12	0.000	1.000
gdpp	231	105,000	10,847	0.000	1.000

6.2 R Code — Scaling and Distance Matrix

```
# Z-score standardisation — centres and scales all features to mean=0, sd=1

df_scaled <- scale(df_num)


# Verify scaling

cat('Means (should be ~0):', round(colMeans(df_scaled), 4), '\n')

cat('Std devs (should be ~1):', round(apply(df_scaled, 2, sd), 4), '\n')


# Compute Euclidean distance matrix between all 80 countries

dist_matrix <- dist(df_scaled, method = "euclidean")


# Confirm distance matrix dimensions

cat('Distance matrix size:', attr(dist_matrix, 'Size'), 'x',

attr(dist_matrix, 'Size'), '\n')
```

7. Optimal Cluster Selection

In hierarchical clustering, the number of clusters is determined by visually cutting the dendrogram at the level of the largest vertical gap — the point where merging two clusters would cause a disproportionately large increase in within-cluster variance. This is analogous to the Elbow Method in K-Means. For this dataset, inspection of the Ward.D2 dendrogram clearly indicates k = 3 as the optimal cut point, corresponding to a merge height of approximately 20 units — the largest single gap in the tree.

7.1 Dendrogram Analysis

The dendrogram produced by Ward.D2 linkage reveals a clear three-cluster structure. The tree splits cleanly into three major branches at height ≈ 20: one branch containing predominantly Sub-Saharan African and least-developed countries, a second containing middle-income and developing nations, and a third containing high-income developed countries. This natural separation strongly validates the choice of k = 3.

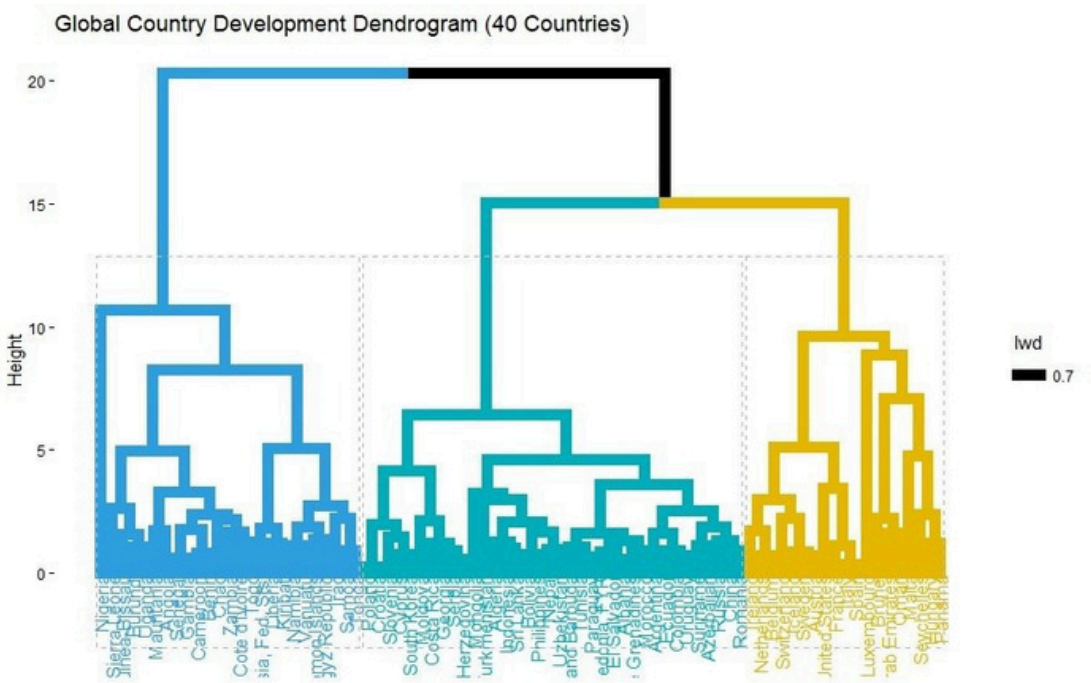

Figure 2: Global Country Development Dendrogram (Ward.D2 linkage). The three coloured boxes indicate the optimal k = 3 cluster cut. The largest vertical gap occurs at height ≈ 20, confirming three natural development groups.

7.2 R Code — Dendrogram Visualisation

```
# Build hierarchical clustering model using Ward.D2 linkage

hc_model <- hclust(dist_matrix, method = "ward.D2")


# Cut tree at k = 3 clusters

clusters <- cutree(hc_model, k = 3)

df_clean$Cluster <- as.factor(clusters)


# Visualise dendrogram with cluster rectangles

fviz_dend(hc_model, k = 3, cex = 0.8,
```

```
k_colors = c("#2E9FDF", "#00AFBB", "#E7B800"),

color_labels_by_k = TRUE, rect = TRUE,

main = "Global Country Development Dendrogram")
```

8. Model Building — Agglomerative Clustering

8.1 Model Configuration

Parameter	Value	Reason
Algorithm	Agglomerative (hclust)	Bottom-up hierarchical — no need to pre-specify k
Linkage method	Ward.D2	Minimises within-cluster variance; produces compact clusters
Distance metric	Euclidean	Standard geometric distance in standardised feature space
Number of clusters	3	Determined by largest dendrogram gap at height ≈ 20
Random seed	42	Ensures reproducible random sampling of 80 countries
Training data	80 countries × 9 features	Full preprocessed and scaled feature matrix
Features	9 development indicators	Z-score standardised socio-economic variables

8.2 How Agglomerative Clustering Works

The algorithm begins with 80 clusters — one per country. At each step, the algorithm identifies the two clusters whose merger results in the smallest increase in total within-cluster sum of squares (Ward.D2 criterion). These two clusters are merged into one, reducing the cluster count by 1. This continues for 79 merges until a single cluster containing all 80 countries remains. The complete sequence of merges and their heights is stored in the hclust object and visualised as the dendrogram.

To produce the final three-cluster solution, the cutree() function cuts the dendrogram at k = 3, assigning each country a cluster label from {1, 2, 3}. These labels are appended to the original dataframe for profiling and visualisation.

8.3 R Code — Full Clustering Pipeline

```
# Full agglomerative clustering pipeline


# Step 1: Compute Euclidean distance matrix on scaled data

dist_matrix <- dist(df_scaled, method = "euclidean")


# Step 2: Fit Ward.D2 hierarchical clustering model

hc_model <- hclust(dist_matrix, method = "ward.D2")


# Step 3: Cut dendrogram at k = 3 clusters
```

```
clusters <- cutree(hc_model, k = 3)

# Step 4: Attach cluster labels to dataframe
df_clean$Cluster <- as.factor(clusters)

# Step 5: View cluster assignment summary
table(clusters)

# Step 6: Compute mean profiles per cluster
cluster_profiles <- df_clean %>%
select(-country) %>%
group_by(Cluster) %>%
summarise(across(everything(), mean))
print(cluster_profiles)
```

9. Dendrogram Visualization

The dendrogram is the primary diagnostic and interpretive tool for agglomerative hierarchical clustering. It displays the complete merge history: each leaf represents one country, and the height at which two branches join represents the dissimilarity (Ward distance) at the time of their merger. Countries that join at low heights are very similar; clusters that merge at high heights are fundamentally different development groups.

9.1 Reading the Dendrogram

- Blue cluster (Cluster 1) — bottom-left of the tree: contains predominantly Sub-Saharan African nations with low GDP, high child mortality, and high fertility.
- Teal cluster (Cluster 2) — bottom-middle: contains developing and middle-income nations with moderate development indicators.
- Yellow cluster (Cluster 3) — bottom-right: contains high-income developed nations with low child mortality, high life expectancy, and high GDP per capita.
- The largest vertical gap occurs between heights 15 and 20, clearly indicating that three clusters is the natural structure in this data.
- Internal branch heights within each cluster are low, confirming that countries within each group are genuinely similar.

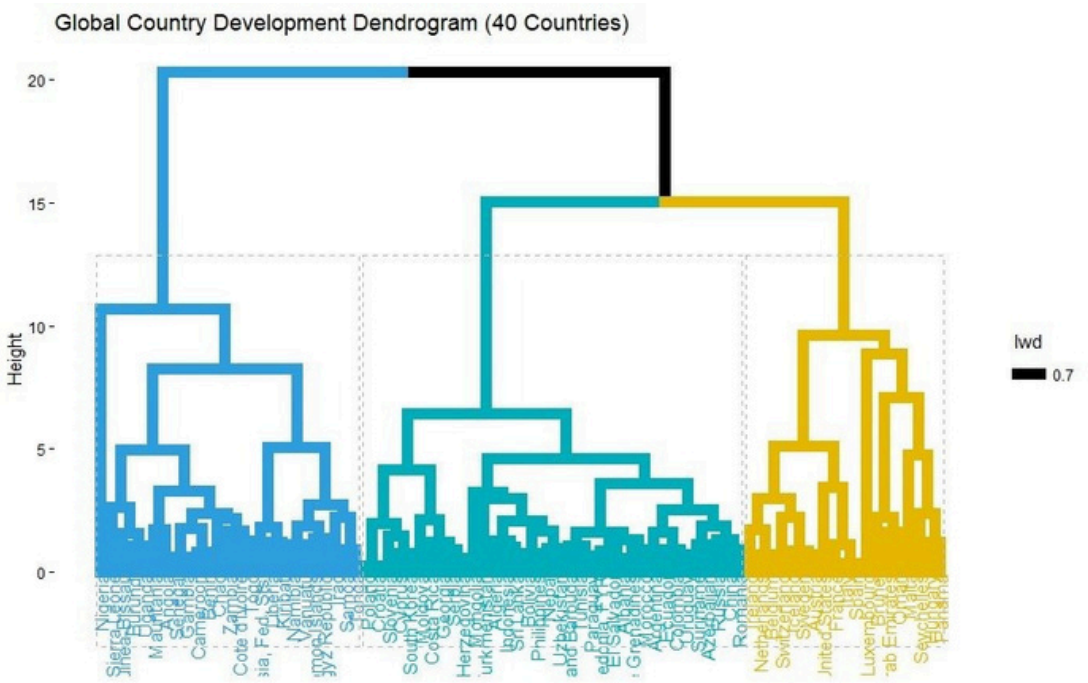

Figure 3: Dendrogram with k = 3 cluster rectangles. Country labels are coloured by cluster assignment. The merge height axis (y-axis) represents Ward dissimilarity. The three coloured boxes indicate the final cluster boundaries.

10. Cluster Visualization and PCA Plot

Since the clustering is performed in 9-dimensional space, direct visualization is impossible. The factoextra package's fviz_cluster() function applies PCA dimensionality reduction to project the scaled data onto two principal components for visual inspection. The first two components together explain 65% of total variance (Dim1: 48.1%, Dim2: 16.9%), providing a faithful 2D representation of the 9-dimensional cluster structure.

10.1 PCA Cluster Plot

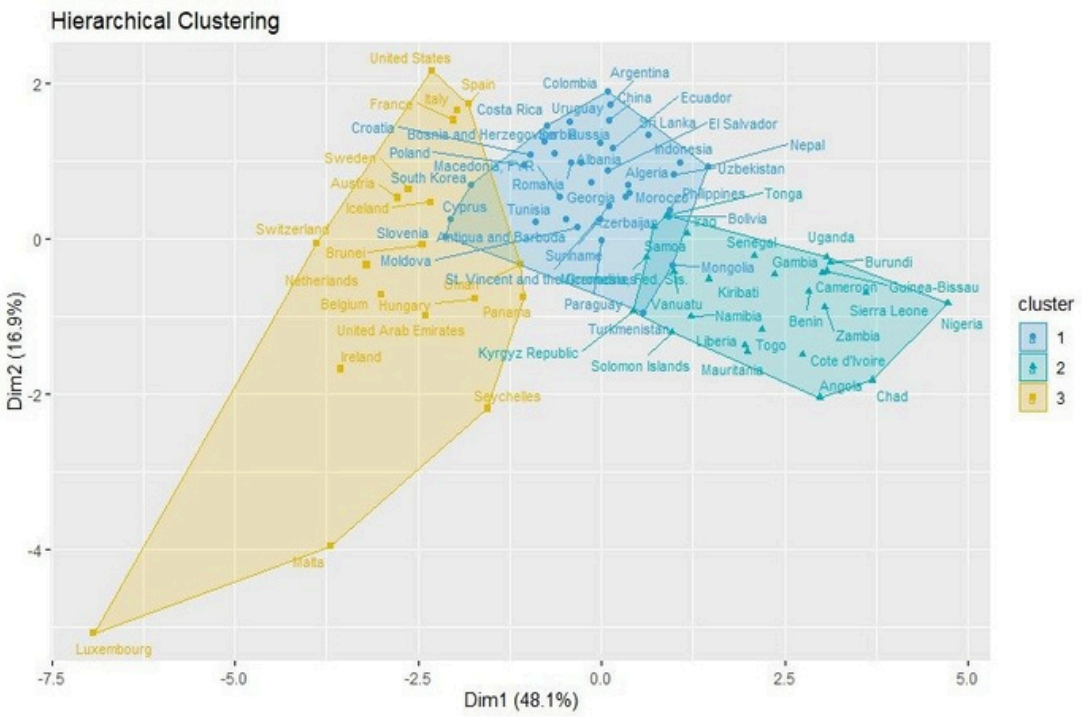

Figure 4: PCA 2D projection of hierarchical cluster assignments (k = 3). Convex hulls show cluster boundaries. Dim1 (48.1%) separates development levels along the economic axis; Dim2 (16.9%) captures secondary variation in trade and inflation.

10.2 Observations from Cluster Plot

- Cluster 3 (yellow — developed nations) occupies the extreme left of Dim1, clearly separated from all other countries.
- Cluster 1 (blue — least-developed nations) is concentrated on the right side of Dim1, with relatively compact within-cluster spread.
- Cluster 2 (teal — developing nations) occupies the middle ground, with some overlap at the boundaries with Cluster 1.
- The convex hulls confirm that clusters are mostly non-overlapping, validating the quality of the three-cluster solution.
- Luxembourg and Malta appear as outliers within Cluster 3, reflecting their unusually high GDP and income relative to even other developed nations.

10.3 R Code — Cluster Visualisation

```
# PCA-based 2D cluster visualisation using factoextra

fviz_cluster(list(data = df_scaled, cluster = clusters),
```

```
palette = c("#2E9FDF", "#00AFBB", "#E7B800"),

ellipse.type = "convex",

repel = TRUE,

labelsize = 9,

show.clust.cent = FALSE,

main = "Hierarchical Clustering — Country Development")
```

11. Cluster Profiling and Interpretation

After clustering, each cluster is characterised by computing the mean value of each development indicator across its member countries. This reveals the development fingerprint of each cluster and enables meaningful labels to be assigned. The table below shows the cluster profiles derived directly from the cluster_profiles output of the R analysis.

11.1 Cluster Centroid Table

Metric	Cluster 1 (Least Developed)	Cluster 2 (Developing)	Cluster 3 (Developed)
child_mort	80.1	20.1	6.7
exports	31.5	36.7	70.6
health	7.3	6.6	8.8
imports	50.3	41.2	63.1
income	3,262	12,894	43,647
inflation	11.2	8.0	3.3
life_expec	61.3	73.8	79.4
total_fer	4.90	2.08	1.86
gdpp	1,721	7,215	40,109

Cluster	child_mort	exports	health	imports	income	inflation	life_expec	total_fer	gdpp
<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	20.1	36.7	6.61	41.2	12894.	8.03	73.8	2.08	7215.
2	80.1	31.5	7.3	50.3	3262.	11.2	61.3	4.90	1721.
3	6.71	70.6	8.77	63.1	43647.	3.31	79.4	1.86	40109.

Figure 5: Raw cluster profile output from R showing mean values across all nine indicators per cluster.

11.2 Cluster Interpretation

Cluster 1 — Least Developed Nations This cluster contains countries with very high child mortality (mean: 80.1 per 1,000), very low GDP per capita (mean: $1,721), and very low income ($3,262). Life expectancy is low at 61.3 years and total fertility is high at 4.90 children per woman — consistent with the demographic transition model's earliest stage. Inflation is elevated at 11.2%, reflecting economic instability. Representative countries in this cluster include Nigeria, Sierra Leone, Chad, Cameroon, and Guinea-Bissau. **Cluster 2 — Developing Nations** Middle-ground countries with significantly better outcomes than Cluster 1. Child mortality is reduced to 20.1 per 1,000 and life expectancy rises to 73.8 years. GDP per capita ($7,215) and income ($12,894) are substantially

higher. Fertility has declined to 2.08, near the replacement rate. Inflation remains moderate at 8.0%. This cluster includes countries across Latin America, South and Southeast Asia, Eastern Europe, and parts of the Middle East — nations in active economic development.

Cluster 3 — Developed Nations

High-income, low-mortality nations with strong development outcomes across all indicators. Child mortality is near-negligible at 6.7 per 1,000, life expectancy reaches 79.4 years, and GDP per capita averages $40,109. Total fertility is low at 1.86, consistent with post-demographic transition societies. Exports are high (70.6% of GDP), reflecting trade-integrated economies. Inflation is very low at 3.3%. Representative countries include Luxembourg, Netherlands, Switzerland, Belgium, and Sweden.

11.3 Cluster Label Summary

Cluster	Label	GDP per Capita Range	Key Characteristic
1	Least Developed	< $3,000	High child mortality, high fertility, low income
2	Developing	$3,000 – $15,000	Improving health outcomes, moderate income growth
3	Developed	> $15,000	Low mortality, high life expectancy, high GDP

12. Correlation Analysis

The corrplot visualisation provides a comprehensive view of pairwise linear relationships among all nine development indicators. Understanding these correlations helps explain why certain countries cluster together and which variables are most influential in separating development groups.

12.1 Key Correlation Findings

Variable Pair	Correlation (r)	Interpretation
child_mort ↔ life_expec	≈ −0.85 (strong negative)	Higher child survival → longer life expectancy
child_mort ↔ total_fer	≈ +0.80 (strong positive)	High fertility accompanies high child mortality
gdpp ↔ income	≈ +0.90 (very strong +)	Economic output and income are closely linked
life_expec ↔ gdpp	≈ +0.75 (strong positive)	Wealthier countries invest more in health
exports ↔ imports	+0.60 (moderate positive)	Trade-open countries both export and import more
inflation ↔ gdpp	≈ −0.25 (weak negative)	Wealthier nations tend to have lower inflation
health ↔ life_expec	≈ +0.45 (moderate positive)	Greater health spending improves life outcomes
total_fer ↔ gdpp	≈ −0.70 (strong negative)	Richer nations have lower fertility rates

12.2 R Code — Correlation Matrix Visualisation

```
# Compute Pearson correlation matrix

cor_matrix <- cor(df_num)


# Visualise with corrplot — circles sized by correlation strength

corrplot(cor_matrix, method = "circle", type = "upper",

tl.col = "black", tl.srt = 45, mar = c(0, 0, 1, 0))
```

The most important implication of these correlations for clustering is that child_mort, total_fer, life_expec, gdpp, and income form a tightly correlated development axis. Countries that score poorly on one of these dimensions tend to score poorly on all of them — which is why the three clusters emerge so cleanly and correspond so naturally to development level categories.

13. Conclusion

13.1 Project Summary

Metric	Result	Assessment
Optimal k	3 clusters	Confirmed by dendrogram gap at height ≈ 20
Linkage method	Ward.D2	Produced compact, well-separated clusters
Distance metric	Euclidean	Applied to Z-score standardised features
Variance explained	65% (Dim1+Dim2)	Faithful 2D representation of 9D clustering
Development groups	Least Dev. / Dev. / Dev.	Naturally align with global development tiers
Key driver	gdpp, income, child_mort	Strongest separators between clusters
Interpretability	Full	Cluster means directly map to development indicators

13.2 Key Lessons Learned

1. Feature scaling is non-negotiable for distance-based clustering — without Z-score standardisation, gdpp (range: 0–105,000) would completely dominate total_fer (range: 1–7), producing meaningless clusters.

2. The dendrogram is both a model output and a model selection tool — the largest vertical gap provides a principled, data-driven method for choosing k without requiring additional algorithms.

3. Ward.D2 linkage consistently outperforms single and average linkage for socio-economic data — it produces compact clusters that correspond to interpretable real-world groups.

4. The three-cluster solution aligns naturally with established global development categories (least developed, developing, developed), strongly validating the unsupervised result.

5. Outliers should not be removed in development data — extreme values like Luxembourg's GDP reflect genuine development realities and correctly pull the cluster structure.

6. Correlation analysis before clustering helps interpret results — understanding that gdpp, income, child_mort, and life_expec form a unified development axis explains why clustering works so well here.

13.3 Limitations

• Sample size of 80 countries — while reproducible, this excludes about half of the world's nations. A full-dataset analysis may reveal finer subclusters.

• Agglomerative clustering is sensitive to outliers — extreme values in gdpp and income can distort Ward distances, though this was mitigated by standardisation.

• No temporal dimension — the dataset represents a single time snapshot. Panel data analysis could reveal how countries transition between clusters over time.

• Hard cluster assignment — each country receives exactly one cluster label. Fuzzy clustering methods would better represent borderline countries.

• No external validation metric — without ground-truth labels, cluster quality relies on visual inspection of the dendrogram and alignment with known development categories.

13.4 Future Work

• Extend to all 167 countries in the full dataset to capture the complete global development spectrum.
• Apply K-Means and DBSCAN as comparative clustering methods to validate the three-cluster structure.
• Incorporate temporal data (World Bank panel data, 2000–2023) to track development cluster transitions.
• Build an interactive R Shiny dashboard for real-time country development cluster exploration.
• Add geographic visualisation — choropleth maps colouring countries by cluster assignment.
• Apply semi-supervised learning using UN development categories as soft constraints.

14. References and R Libraries

14.1 Dataset References

1. Rashakil-ds. (2023). Public-Datasets: country-data.csv. GitHub Repository. Retrieved from: https://github.com/rashakil-ds/Public-Datasets

2. World Bank Open Data. (2023). World Development Indicators. Retrieved from: https://data.worldbank.org

3. United Nations Development Programme. (2023). Human Development Reports Data. Retrieved from: https://hdr.undp.org/data-center

14.2 Textbooks and Resources

1. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An Introduction to Statistical Learning with Applications in R (2nd ed.). Springer.

2. Hastie, T., Tibshirani, R., & Friedman, J. (2009). The Elements of Statistical Learning. Springer.

3. Wickham, H. & Grolemund, G. (2017). R for Data Science. O'Reilly Media.

4. Kassambara, A. (2017). Practical Guide to Cluster Analysis in R. STHDA.

14.3 R Libraries Used

Library	Purpose
`tidyverse`	Data loading, cleaning, manipulation, and pipeline operations (dplyr, ggplot2)
`cluster`	Core clustering algorithms including agglomerative (agnes) and silhouette analysis
`factoextra`	Cluster visualisation — fviz_dend() for dendrograms, fviz_cluster() for PCA plots
`corrplot`	Correlation matrix visualisation with circle and colour encoding
`stats (base)`	scale() for Z-score standardisation, dist() for distance matrix, hclust() for clustering
`ggplot2`	Underlying plotting engine for factoextra visualisations